

Agentic Customer 360: Proactive Intervention Desk

Mid-Term Progress & Future Execution Report

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Track: Natural Language Processing (NLP) — **Repository:** github.com/0mM1shra/inter_iit_nlp_mid

1. Problem Domain Insights & Key Challenges

Traditional enterprise CRM platforms rely on passive dashboards requiring manual account manager monitoring. Transitioning to an **Ambient Agentic Customer 360 Desk** introduces three core domain challenges:

- **Ambient Execution vs. Prompt-Based Chatbots:** System must operate as an asynchronous background engine, continuously triggered by multi-modal event streams or scheduled account checks rather than waiting for human prompts.
- **Early Lead Time vs. Premature Overreaction:** The system must infer subtle behavioral shifts early (e.g., identifying pre-churn savings withdrawal) while suppressing premature, low-confidence interventions on noise.
- **Red Herring Isolation:** Real-world financial telemetry contains high-variance noise (e.g., tuition transfers, vacation refunds, tax deposits). The system must correlate multi-domain signals over time to isolate real signals.

2. Dataset & Event Stream Learnings

Deep analysis of the multi-source event stream package and practice scenario timelines yielded key structural insights:

- **Multi-Source Schema Envelope:** Payload ingestion normalizes streams across `card_payments`, `ach_wire`, `core_banking_led`, `trading_brokerage`, `loan_kyc`, `web_app_events`, and `support_logs`.
- **Out-of-Order Watermarking:** Ingested events are replayed via an event-time watermark queue (`event_time` vs. `ingestion_time`) to prevent late-arriving packets from corrupting 30-day windowed rolling rollups.
- **Cross-Domain Signal Correlation:** Evaluated ground-truth timelines (*Medical Hardship*, *New Child Event*, *Churn Risk*) confirm that high-confidence state inference requires correlating signals across KYC updates, transaction anomalies, and support ticket sentiment.

3. Current Architectural Progress Achieved

- **Per-Customer Shared State Board:** Designed a structured blackboard scoped per customer ID. Swarm agents (`UsageAgent`, `SupportAgent`, `TxnAgent`, `KYCAgent`) write structured conclusions to eliminate multi-tenant memory leakage.
- **3-Tiered Memory Architecture:** Formulated Working Memory (session context), Episodic Memory (decay-weighted historical event RAG), and Semantic Memory (product/policy RAG).
- **Multi-Agent Debate & Refiner Topologies:** Implemented debate protocols for reconciling contradictory swarm reads (e.g., tax deposit vs. app drop) and a Critique-Refiner pass for tone/compliance checking.
- **Safety Rails & Calibrated HITL:** Integrated hardcoded regex guardrails for instant legal/AML threat bypass, data-layer PII redaction, and citation-backed human-in-the-loop (HITL) checkpoints.
- **Automated Evaluation Harness:** Built a Python scoring harness (`evaluate_scenarios.py`) measuring inference accuracy, action precision, HITL calibration, and false-positive avoidance against ground-truth benchmarks.

4. Future Execution Roadmap

- **Phase 1: End-to-End Stream Pipeline Implementation:** Stitching real-time stream replayers, state board storage, and parallel swarm workers into a unified runtime engine.
- **Phase 2: Synthesis Engine & Multi-Agent Debate Layer:** Finalizing LLM prompt chains for life-phase inference, dynamic hypothesis weighting, and automated contradiction resolution.
- **Phase 3: Interactive HITL Dashboard & Audit Logging:** Developing a lightweight human review interface featuring citation-backed "Ask Why" explanations and OpenTelemetry trace logging.
- **Phase 4: Full Scenario Benchmarking & Final Deliverables:** Executing end-to-end evaluation across all stream scenarios, optimizing lead times, and compiling the final 3-page solution document and GitHub repository.